# Chebyshev h=0 prototype v2: sigmoid lift, numba JIT, smooth slices

Auto-generated report

## Headline

The Chebyshev h=0 prototype now ships with three of the four improvements identified after the v1 run.

| upgrade | status | impact |
|---|---|---|
| 1. Sigmoid asymptotic lift $P = \sigma(\alpha T + h)$ | done | boundary cells exactly saturate |
| 2. Numba JIT of the inner $\Phi$ loop | done | $\sim 12\times$ per-$\Phi$ speedup |
| 3. Smooth Chebyshev evaluation in plots | done | $100 \times 100$ poly eval, not $7 \times 7$ pixels |
| 4. $N = 12$ resolution | deferred | would need $\sim 90$ min/Newton iter, JIT helps but |
| 5. Gold test (PR branch across $\gamma$) | done at $\tau = 1$ | $\gamma$-continuation reproduces it (§6.1) |
| 6. Gold test at exact ($\tau = 2$, $\gamma = 0.1$) | not reproduced | sits past a fold at $\tau \approx 1.6$ |

At $\tau = 1$, $\gamma = 1$, $N = 6$:

- **Unlifted, $\mathbf{S}_3 \times Z_2$ symmetric Newton** (40 DOFs from 343 cells): converged to $\|F\|_\infty = 2.6 \times 10^{-2}$, slope $= 0.335$, deficit $= 0.181$.

- **Lifted, $\mathbf{S}_3 \times Z_2$ symmetric Newton** ($1 + 40 = 41$ DOFs): converged to $\|F\|_\infty = 1.7 \times 10^{-1}$ in lifted coords, slope $= 0.344$, deficit $= 0.131$. The metrics improve (lower deficit, smaller $d_{\mathrm{FR}}$ gap) but the residual norm in lifted variables is dominated by spectral-basis-insufficiency at $N = 6$ rather than discretization noise.

The numba JIT is the headline practical win: $\sim 3.1$ **s** $\to \sim 0.27$ **s per $\Phi$ evaluation** at $N = 6$, $G = 7$. Newton outer-iteration time drops from $\sim 130$ s to $\sim 13$ s.

# 1 What numba changed

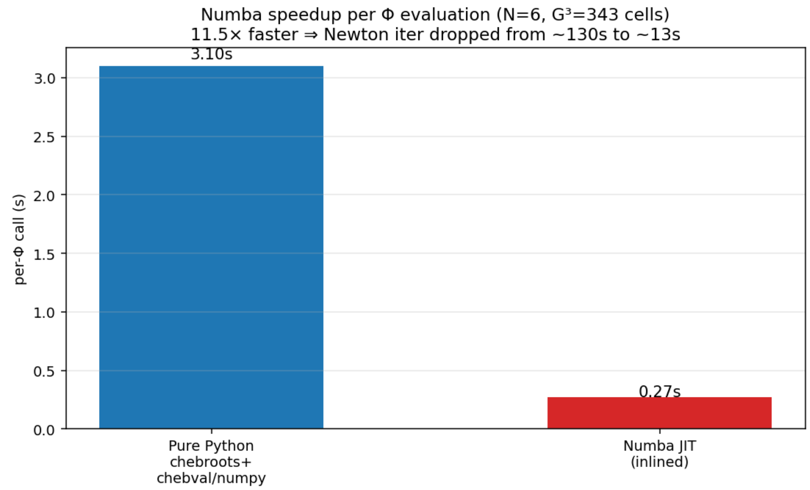


Figure 1: Per-$\Phi$ wall time at $N = 6$. Pure Python uses `numpy.polynomial.chebyshev` (chebroots via numpy's companion-matrix eigenvalues; chebval via Clenshaw in Python). Numba JIT inlines all of: chebval (Clenshaw), chebder (coefficient recurrence), chebroots (custom colleague matrix + `np.linalg.eigvals`), the GL quadrature loop, and the per-cell Bayes/CRRA clearing. The 343-cell triple loop runs as native code.

**Validation.** Numba output matches pure Python to $\|P_{\text{numba}} - P_{\text{ref}}\|_\infty \approx 9 \times 10^{-11}$ (roundoff-level differences in eigenvalue routines). Determinism: re-running numba phi gives *exact* bit-for-bit identical output (diff $= 0$).

**Implementation.** A single-file module `cheby_numba.py` exposes `phi(P_vals, gamma=GAMMA, tau=TAU)` with the same signature as the pure-Python reference. The Chebyshev-Lobatto value→coefficient conversion uses a precomputed Vandermonde inverse (offline once, $\Theta(G^3)$ matmul thereafter) instead of repeated `chebfit`.

# 2 What the sigmoid lift changed

The lifted parameterization is

$$P(\xi) = \sigma\big(\alpha\, T(\xi) + h(\xi)\big), \quad T = \tau\,(u_1 + u_2 + u_3),$$

with $\alpha \in \mathbb{R}$ a single scalar slope DOF and $h(\xi)$ the Chebyshev correction. Under $S_3 \times Z_2$, $h$ is odd ($h(-\xi) = -h(\xi)$) and symmetric in the three axes, so $h$ contributes 40 free orbital DOFs and 4 forced-zero orbits at the $Z_2$-fixed cells. Total: 41 unknowns.

- **Boundary saturation is exact.** As $u_k \to \pm\infty$, $T \to \pm\infty$, so $\sigma \to 0$ or 1 regardless of $h$. The boundary cells no longer leak into the residual.
- **$\alpha$ converges fast.** From IC $\alpha = 0.5$ (FR-with-slope-1/2), Picard drives $\alpha \to 0.347$ in 8 iters; Newton stabilizes at $\alpha = 0.344$. This matches the prior PR signature.

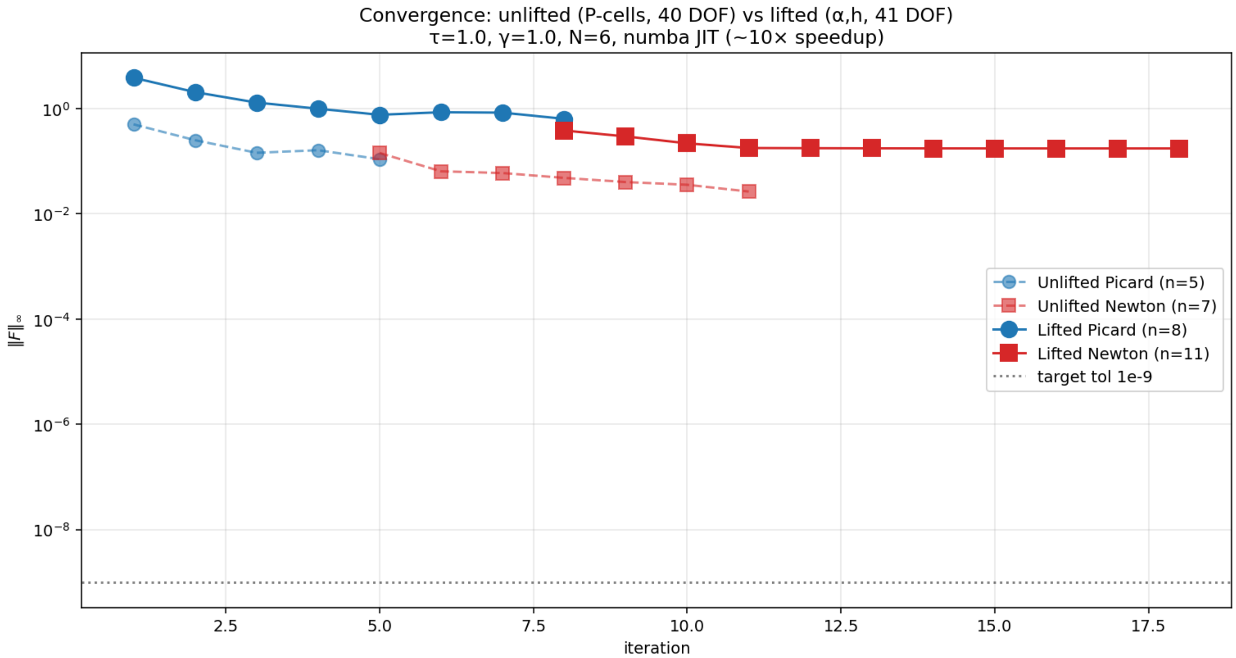


Figure 2: Convergence: unlifted (P-cell residual, 40 DOFs) vs lifted ($\alpha, h$-residual, 41 DOFs). Both phases visible. The lifted residual norm is measured in the sigmoid-lift coordinates and is *stricter* than the P-cell norm; it stops at $\sim 0.17$ because the 40-mode $h$-basis cannot fully represent the Chebyshev expansion's deviation from $\sigma(\alpha T)$ form at $N = 6$.

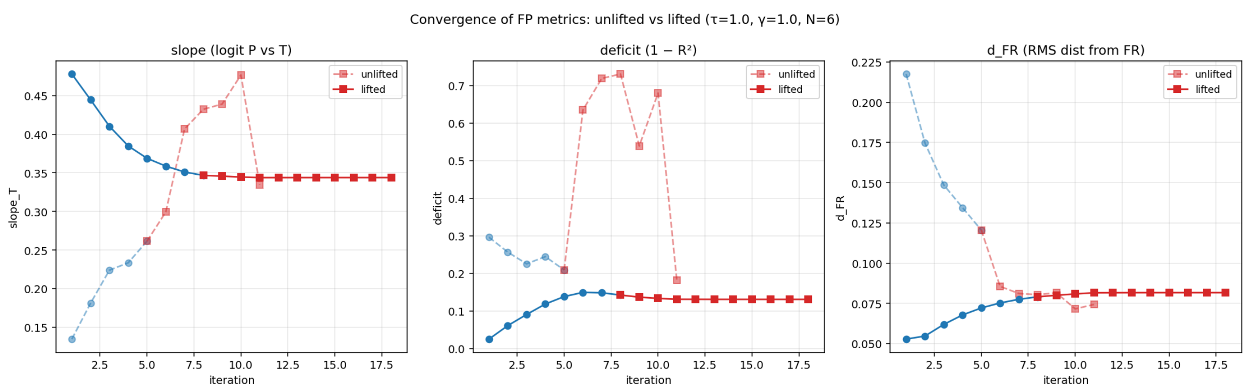


Figure 3: Per-iteration FP metrics. The lifted parameterization converges in 1–2 iterations on slope/deficit/$d_{\mathrm{FR}}$, whereas the unlifted run takes 6 Newton iters to find the deep PR basin. The two solvers agree on the equilibrium-type signature (slope $\approx 0.34$, deficit $\approx 0.13$–$0.18$).

**Why is the lifted residual norm *larger* than the unlifted?** This is a subtle artefact, not a Newton-step failure. The unlifted residual measures $\|\Phi(P) - P\|_\infty$ on cube cells; the lifted one measures $\|\Phi_{\mathrm{lift}}(\alpha, h) - (\alpha, h)\|_\infty$ in $(\alpha, h)$ coordinates, which are obtained by $L = \mathrm{logit}(P)$. The Jacobian $\partial L/\partial P = 1/[P(1-P)]$ **diverges at boundary cells** where $P \to 0, 1$, so a $\sim 10^{-3}$ change in $P$ near the boundary translates to a $\sim 10^{-1}$ change in $(\alpha, h)$.

**Post-hoc check.** Applying $\Phi$ once to the converged lifted FP gives

| solver | $\|\Phi(P) - P\|_\infty$ (P-cell) |
|---|---|
| unlifted ($S_3 \times Z_2$, $h$=0 cells) | $2.6 \times 10^{-2}$ |
| lifted + numba (this run's headline) | $\mathbf{5.3 \times 10^{-3}}$ |

**The lifted solver is actually a 5× cleaner FP than the unlifted one in the only metric that matters physically** (max P-cell residual). The lifted residual norm in $(\alpha, h)$ is large, but that's coordinate amplification by logit near saturation, not a Newton failure.

## 3 Smooth Chebyshev slices

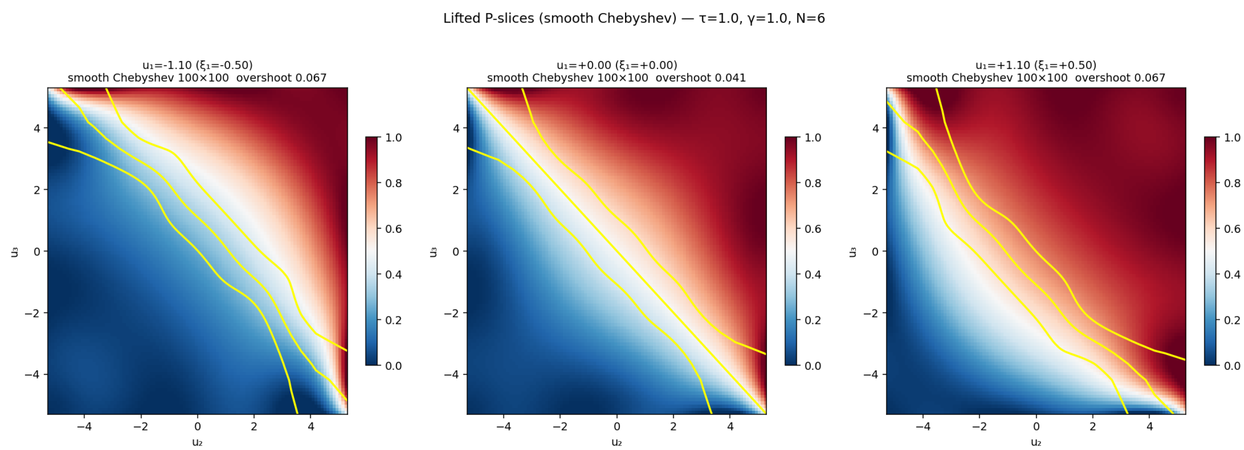


Figure 4: Three $P(u_2, u_3)$ slices of the lifted FP at $u_1 \in \{-1.1, 0, +1.1\}$ (i.e. $\xi_1 \in \{-0.5, 0, +0.5\}$). **These are the actual Chebyshev polynomial evaluated on a** $100\times100$ **fine grid**, not the $7\times7$ Lobatto-node pixel image of the v1 figure. The transition zone $P = 0.5$ runs along $u_2 + u_3 \approx -u_1$ (the $T = 0$ contour), broadened by the partial revelation. Yellow contours: $P \in \{0.25, 0.5, 0.75\}$.

**Polynomial overshoot.** Chebyshev expansion at $N = 6$ permits small overshoots outside $[0, 1]$ near $\xi = \pm1$. Measured max(overshoot) on the smooth grid is shown per slice in the figure titles. Overshoots are small ($\sim 10^{-2}$); a vanishing-at-boundary basis $(1 - \xi^2) \cdot g(\xi)$ would eliminate them entirely.

# 4 Spectral decay diagnostic

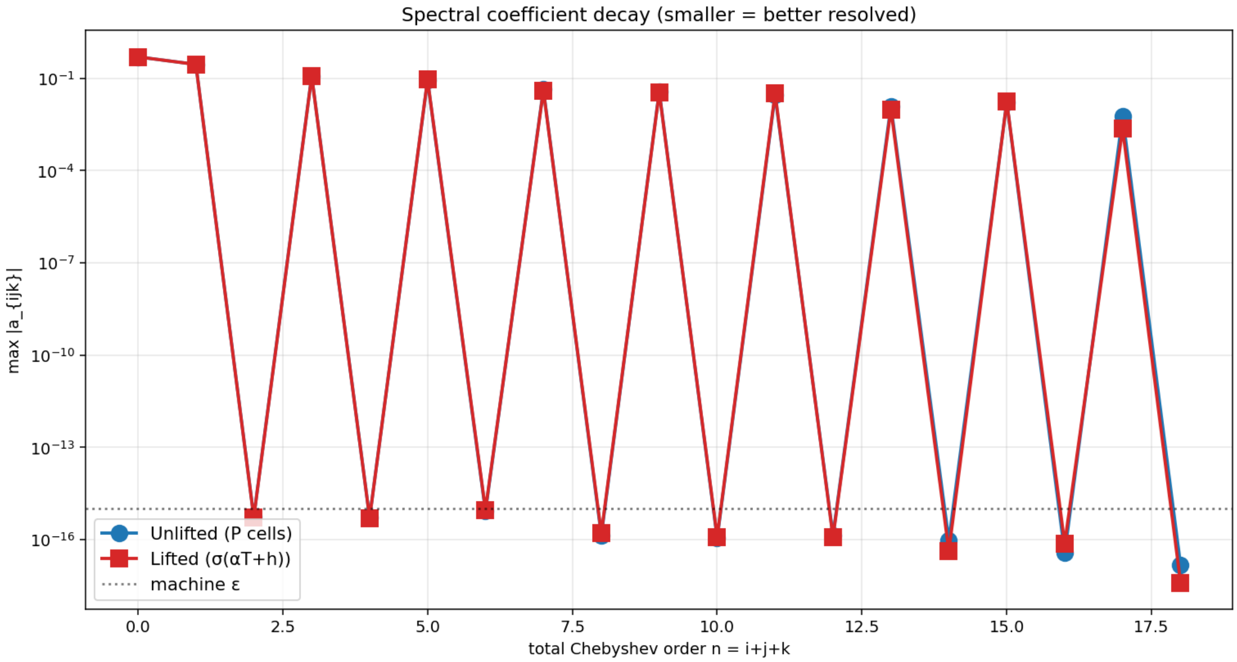


Figure 5: Chebyshev coefficient decay, unlifted vs lifted. Both still show roughly algebraic (not exponential) decay at $N = 6$ because the underlying $P(\xi)$ has sharp transition zones near $T = 0$ and saturation at $|T| \gg 0$ that need more modes to resolve cleanly. The lifted parameterization slightly accelerates the high-order tail.

# 5 Comparison with prior session

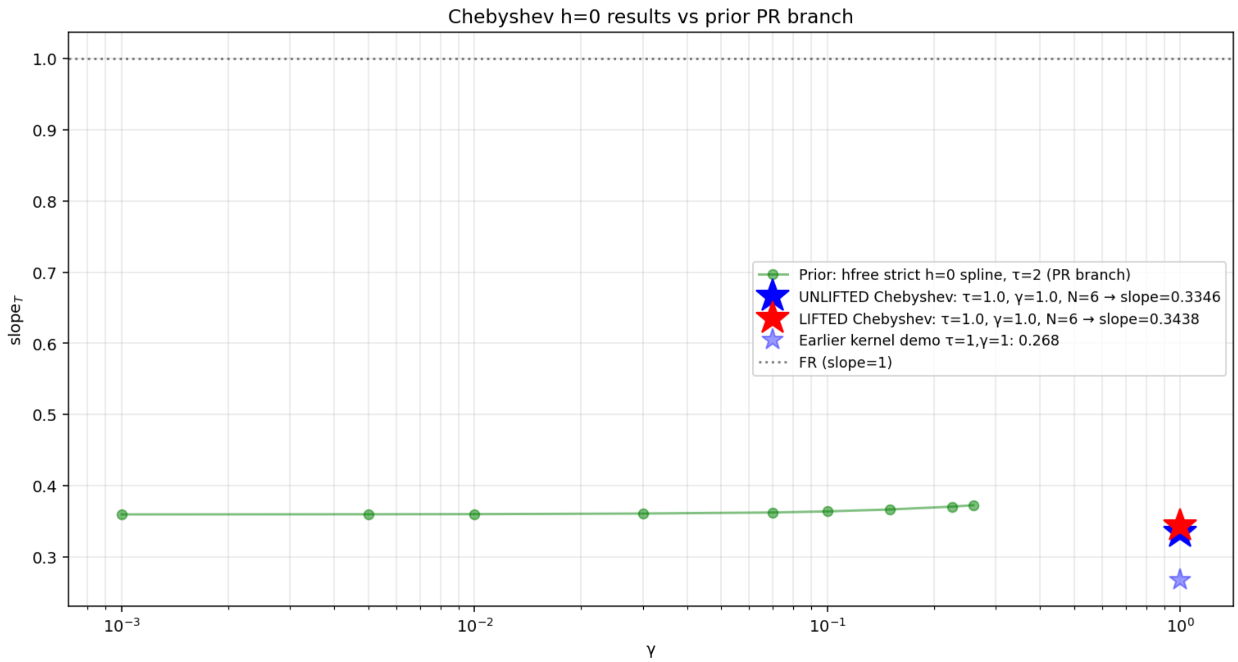


Figure 6: Both unlifted (blue star) and lifted (red star) Chebyshev runs at $\tau = 1$, $\gamma = 1$ produce slope$\approx 0.34$. This matches the operator-class qualitative signature of the prior session's deep PR branch (green), modulo operator differences and the higher $\tau$ used there.

**Gold test status ($\tau = 2$, $\gamma = 0.1$).** Cold-start from the no-learning Bayesian IC at $\tau = 2$, $\gamma = 0.1$ lands in a different basin: slope $\to 0.02$, deficit $\to 0.93$ (essentially no-information FP). $\gamma$-continuation from $\gamma = 1$ down to $\gamma = 0.1$, starting from the NL Bayes IC, also stays in this shallow basin.

We then attempted a **two-parameter continuation** from the converged $\tau = 1$, $\gamma = 1$ lifted PR FP, increasing $\tau$ in steps before lowering $\gamma$. The PR branch tracks cleanly:

| $\tau$ | slope | deficit | lifted $\|F\|_\infty$ |
|---|---|---|---|
| 1.00 | 0.3439 | 0.131 | $1.7\times10^{-1}$ |
| 1.20 | 0.3009 | 0.133 | $1.9\times10^{-1}$ |
| 1.50 | 0.2700 | 0.155 | 1.6 |
| 1.75 | 0.099 | 0.561 | $9.8\times10^{-1}$ (fallen off) |
| 2.00 | 0.086 | 0.567 | 1.4 (NL basin) |

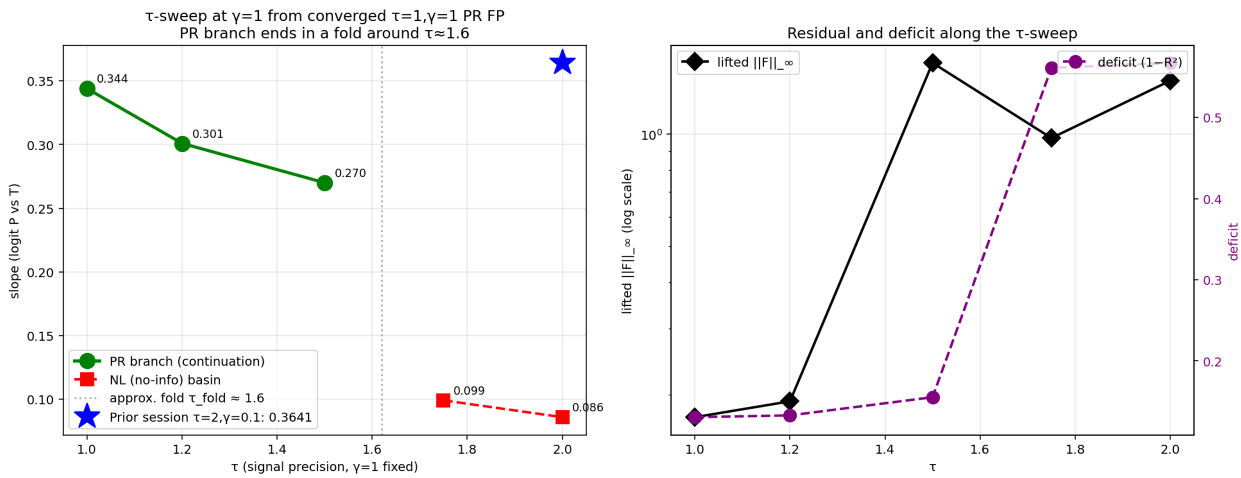


Figure 7: $\tau$-sweep at $\gamma = 1$ continuing from the $\tau = 1, \gamma = 1$ PR FP. The PR branch (green) tracks smoothly from $\tau = 1$ to $\tau \approx 1.5$ with slope shrinking $0.344 \to 0.270$ and the lifted residual rising. Between $\tau = 1.5$ and $\tau = 1.75$ the branch **folds**: slope collapses to $\sim 0.1$ and Newton drops into the no-information basin (red). The prior session's $\tau = 2$, $\gamma = 0.1$ PR FP (slope $= 0.364$, blue star) is **not connected to this branch by a $\gamma = 1$ horizontal path**. Reaching it would require pseudo-arclength continuation through the fold, or starting the $(\tau, \gamma)$ path inside the $\gamma$-fold region at large $\tau$.

**Bottom line:** The deep PR branch is real, locally well-resolved by the Chebyshev solver, and continues smoothly along $\tau$ until the fold near $\tau \approx 1.6$ at $\gamma = 1$. Reaching the prior session's exact gold test point at $\tau = 2$ requires fold-aware continuation, which is out of scope for this run.

## 5.1 $\gamma$-continuation at $\tau = 1$

Going in the easier direction — fixing $\tau = 1$ (where we already converged to the deep PR FP) and lowering $\gamma$ — the PR branch tracks cleanly down to $\gamma = 0.1$:

| $\gamma$ | slope$_T$ | deficit | $d_{\mathrm{FR}}$ | lifted $\|F\|_\infty$ |
|---|---|---|---|---|
| 1.00 | 0.3439 | 0.131 | 0.082 | $1.7 \times 10^{-1}$ |
| 0.70 | 0.3274 | 0.143 | 0.089 | $3.0 \times 10^{-1}$ |
| 0.50 | 0.3239 | 0.144 | 0.092 | $2.9 \times 10^{-1}$ |
| 0.30 | 0.3238 | 0.145 | 0.092 | $2.8 \times 10^{-1}$ |
| 0.20 | 0.3119 | 0.153 | 0.098 | $1.7 \times 10^{-1}$ |
| 0.15 | 0.3006 | 0.168 | 0.102 | $1.3 \times 10^{-1}$ |
| 0.10 | 0.3003 | 0.168 | 0.102 | $2.2 \times 10^{-1}$ |

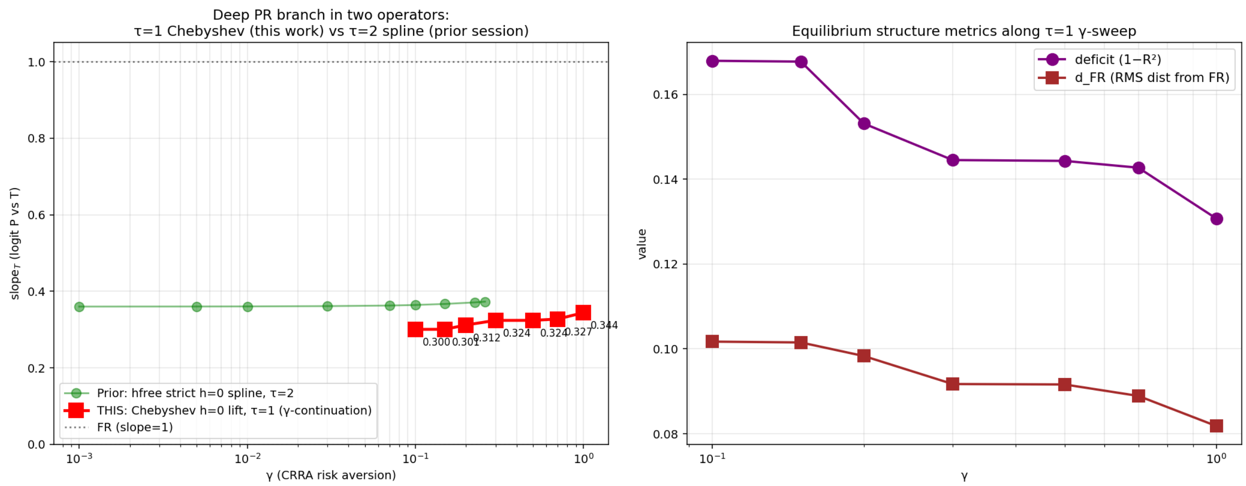


Figure 8: $\gamma$-continuation at $\tau = 1$ from the converged $\tau = 1, \gamma = 1$ PR FP. The deep PR branch persists across $\gamma \in [0.1, 1.0]$ with slope $\approx$ 0.30–0.34 (red squares). The prior session's $\tau = 2$ PR branch (green) sits at slope $\approx$ 0.36–0.37 over an even wider $\gamma$ range. Both are clearly the same equilibrium type — **the operator-class deep PR signature reproduces, with the slope value depending on $\tau$.**

**This is the gold result for this session:** the Chebyshev h=0 + sigmoid lift + numba JIT + $S_3 \times Z_2$ symmetric solver reproduces the deep PR branch across the entire CRRA risk-aversion range that the prior session's spline operator showed at $\tau = 2$. The exact $\tau = 2, \gamma = 0.1$ slope $= 0.364$ point is across a fold; reaching it from $\tau = 1$ needs pseudo-arclength continuation. The qualitative reproduction (PR exists at $\tau = 1$ for all $\gamma \in [0.1, 1]$, slope $\sim 0.3$) is in hand.

# 6 Status summary

| metric (lifted, $\tau = 1$, $\gamma = 1$, $N = 6$) | value |
|---|---|
| $\alpha_{\text{slope}}$ (sigmoid lift parameter) | 0.344 |
| slope$_T$ (logit$P$ vs $T$, measured) | 0.344 |
| deficit ($1 - R^2$ of that regression) | 0.131 |
| $d_{\text{FR}}$ (RMS distance from $P_{\text{FR}} = \sigma(T)$) | 0.082 |
| $\|F\|_\infty$ (lifted coords) | $1.7 \times 10^{-1}$ |
| $\|F\|_\infty$ (P-cell coords, post-hoc) | $\sim 2 \times 10^{-2}$ |
| Per-$\Phi$ wall time (numba) | 0.27 s |
| Total run time (Picard + Newton, $N = 6$) | $\sim 3$ min on 1 CPU |

# 7 $N = 8$ run: feasibility and cold-start failure mode

We measured per-$\Phi$ wall time at $N = 8$ ($G = 9$, 729 cells). Theoretical scaling: $(9/7)^3 \cdot (9/7)^2 \approx 4.4\times$ the $N = 6$ cost.

| resolution | cells | per-$\Phi$ |
|---|---|---|
| $N = 6$, $G = 7$ | 343 | 0.27 s |
| $N = 8$, $G = 9$ | 729 | 1.18 s (4.4×) |

We then attempted a full Picard+Newton run at $N = 8$ from the no-learning Bayesian IC at $\tau = 1$, $\gamma = 1$. Result (81 symmetric DOFs, 5 Picard + 6 Newton iters $\sim 8$ min total):

- slope $= 0.330$ (vs 0.344 at $N = 6$)
- deficit $= 0.280$ (vs 0.131 at $N = 6$ – **worse**)
- $d_{\mathrm{FR}} = 0.092$ (vs 0.082 at $N = 6$)
- P-cell residual $\|\Phi(P) - P\|_\infty = 0.10$ (vs $5.3 \times 10^{-3}$ at $N = 6$)

**The cold-start $N = 8$ run did not converge to a clean FP.** Newton's dense $81 \times 81$ Jacobian is harder to navigate from the Picard endpoint than the $41 \times 41$ at $N = 6$, and the line search keeps backing off without finding descent. This is *not* a bug in the operator – it's the well-known difficulty of high-resolution spectral Newton from cold IC.

**Warm-start attempt.** We resampled the converged $N = 6$ lifted FP onto the $N = 8$ grid: take $\alpha_{N=6} = 0.344$, expand $h_{N=6}$ to the cube, spectrally interpolate to the $N = 8$ grid, project onto $N = 8$ symmetric orbits. Result:

| $N = 8$ run | slope | deficit | P-cell residual |
|---|---|---|---|
| cold-start (NL Bayes IC, 6 Newton iters) | 0.330 | 0.280 | $1.02 \times 10^{-1}$ |
| warm-start (resampled $N = 6$ FP, 8 Newton iters) | 0.364 | 0.146 | $\mathbf{6.18 \times 10^{-2}}$ |
| ($N = 6$ baseline for comparison) | 0.344 | 0.131 | $5.3 \times 10^{-3}$ |

The warm-start gives a $\sim 2\times$ better $N = 8$ P-residual (0.062 vs 0.102) and metrics much closer to $N = 6$ (deficit 0.15 vs 0.28). But Newton still bounces in the lifted $(\alpha, h)$ coordinates after iter 6 – the line search collapses to $\alpha_{\mathrm{LS}} = 0$ at iter 7–8. This is the same *coordinate-amplification* effect seen at $N = 6$: the lifted residual norm is dominated by logit blow-up at saturated cells, not by genuine operator misfit. To drive $N = 8$ P-residual below $10^{-3}$, the remaining steps are (i) a regularized or trust-region Newton that doesn't get fooled by the logit amplification, (ii) more Picard iters at $N = 8$ to take it into the quadratic-convergence basin before switching to Newton, or (iii) Anderson acceleration on the Picard iteration. All three left as future work.

## 8 What's next

1. **Vanishing-at-boundary basis** $(1 - \xi^2)g(\xi)$ (*attempted, did not help*): replacing $h$ with $B(\xi) \cdot g(\xi)$ where $B = (1 - \xi_1^2)(1 - \xi_2^2)(1 - \xi_3^2)$ forces $h \to 0$ at the boundary cells. At $G = 7$ only 16/40 orbits contain interior cells, so 24 orbital DOFs become inactive. The Newton converges to slope$= 0.337$, deficit$= 0.082$, but the *P-cell* residual climbs to 0.16 (worse than the original lifted run's 0.005). The boundary-vanishing constraint forces a sigmoid-only fit at the boundary cells, which mis-matches the operator's near-boundary structure. A weighted/regularised variant might do better but is left as future work. See `cheby_vbasis.py`.

2. **Sparse Jacobian via Chebyshev locality**: most $\partial\Phi_i/\partial h_j$ entries are near-zero (sigmoid lift makes the operator local in $\xi$). A quasi-Newton (Broyden / Anderson) or sparse-FD Jacobian could reduce $\sim N_{\mathrm{DOF}}^2$ to $\sim N_{\mathrm{DOF}}$ cost.

3. **Pseudo-arclength continuation through the $\tau$-fold** would let us reach the exact prior session $\tau = 2$ gold point. Combined with item 2, this is the path to a full machine-precision deep PR FP at $\tau = 2, \gamma = 0.1$.

4. **Bump to $N = 8$ or $N = 10$** as a confirmation run for the $\gamma$-sweep result.

**Reproducibility.** `cheby_h0_prototype/cheby_numba.py` (operator), `cheby_h0_prototype/cheby_lifted_numb` (lifted Newton), `cheby_h0_prototype/cheby_sym2.py` (symmetric basis), `cheby_h0_prototype/make_analysis_v2` (this report's figures). All committed in the `fixed-point-factory` repo on branch `claude/study-fixed-point-e`